Segment-7

- System Implementation - activities.
- Testing method, classification
- Software application testing
- User acceptance testing
- Installation
- Documentation
- Training
- EFSS
- Support information system
- Implementation success factors
- Project close down
- System maintenance, Activities, Types
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- Controlling maintenance request
- Configuration management
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Purpose of system implementation: The purpose of system implementation is to make the new system available to a prepared set of users.

⇒ To convert final physical system specifications into working and reliable software.

⇒ To document work that has been done.

⇒ To provide help for current and future users.

Six activities in system: ① Coding ② Testing ③ Installation ④ Documentation ⑤ Training ⑥ support

Coding- Coding is an act of translation. Physical design specifications are turned into working computer code.

Testing - Testing is basically a process to detect errors in the software product.

⇒ It is performed using various strategies.

⇒ It can be performed in parallel with coding.

Installation - Installation refers to the activities needed to install an information system and successfully convert an organization to using it.

⇒ The current system is replaced by the new system.

Documentation: It is the process of recording information on a reference or operational purposes.

Training: It is the process of identifying necessary skills and knowledge for effective information system work.

Software application testing:

=> A master plan is developed during the analysis phase.
=> During the design phase, unit, system and integration test plans are developed.
=> Actual testing is done during implementation.
=> Test plans provide improved communication.

Testing methods: ① **Unit testing:** Each module is tested alone in an attempt to discover any errors in its code.

② **Integration testing:** It is the process of bringing together all of the modules that a program comprises for testing purposes. It is the process of testing the interface between two software units or modules.

③ **System testing:** It is the process of bringing together all of the programs: programs are typically integrated in a top-down. It is a type of software testing that evaluates the overall functionality.

Test classification:

① **Static test:** It means that the code being tested is not executed.
② **Dynamic test:** It involves execution of the code.
③ **Automated test:** It means that the computer conducts the test.
④ **Manual test:** It means that people complete the test.

⇒ Manual Vs Automated

⇒ Static Vs Dynamic
↓ (Syntax only) ↓ (execution)

Manual testing techniques:

⇒ Inspection (participants examine program for predetable language specific errors)

⇒ Walkthrough (created system development process, also called a structured walkthrough)

⇒ Desk checking (Program code is sequentially executed manually by the reviewer)

Automated testing techniques:

⇒ Syntax checking (To identify syntax errors)

⇒ Unit testing (each module is tested alone, also called module testing)

⇒ Integration testing (bringing together all of the modules, modules are typically integrated in a top-down, incremental fashion).

⇒ System testing (Each module is tested as it is written) Top modules contain many calls to subordinate modules.

⇒ Stub testing (It is a technique used in testing, where modules are written and tested in a top down fashion where few lines of code are used to substitute)

Software application testing (the testing process) –

① system satisfies requirements.

② Testing must be planned.

Test cases: ① Test cases are usually developed by analysts

② Test cases should not be created by the Programmers.

③ Programmers use symbolic debuggers to isolate errors.

User acceptance testing: Alpha testing – use simulated data

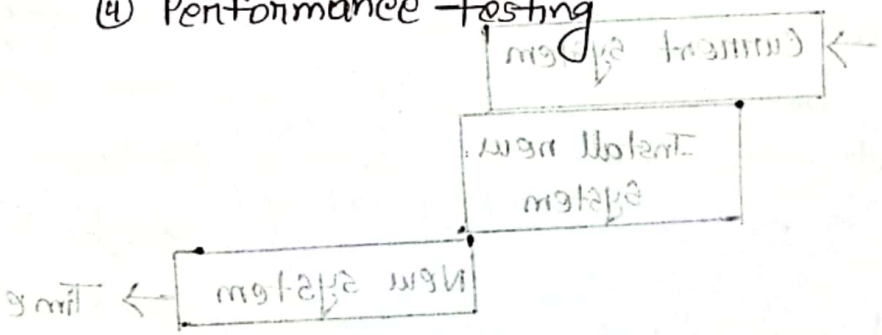
Beta testing – use real data in real user environment

Types of Alpha testing – ① Recovery testing

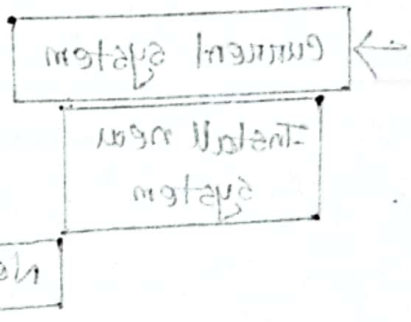
② Security testing

③ Stress testing (tries to break the system)

④ Performance testing



Parallel Installation: Running the old information system and the new one at the same time until management decides the old system can be turned off.



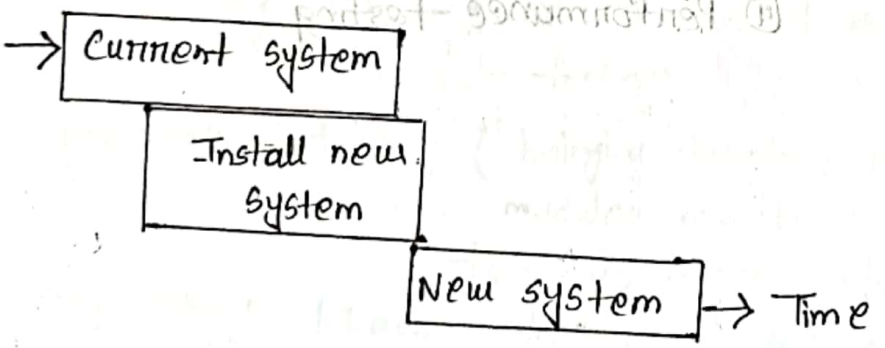
Installation: The organizational process of changing over from the current information system to new one.

⇒ It refers to the activities needed to install an information system and successfully convent an organization to using it.

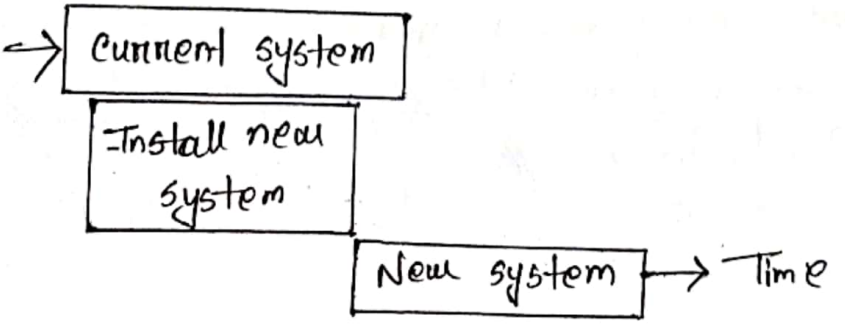
There are 4 approaches of installation. They are-

① Direct installation
② Parallel installation
③ Single location installation
④ Phased installation

Direct installation: changing over from the old information system to a new one by turning off the old system when the new one is turned on.

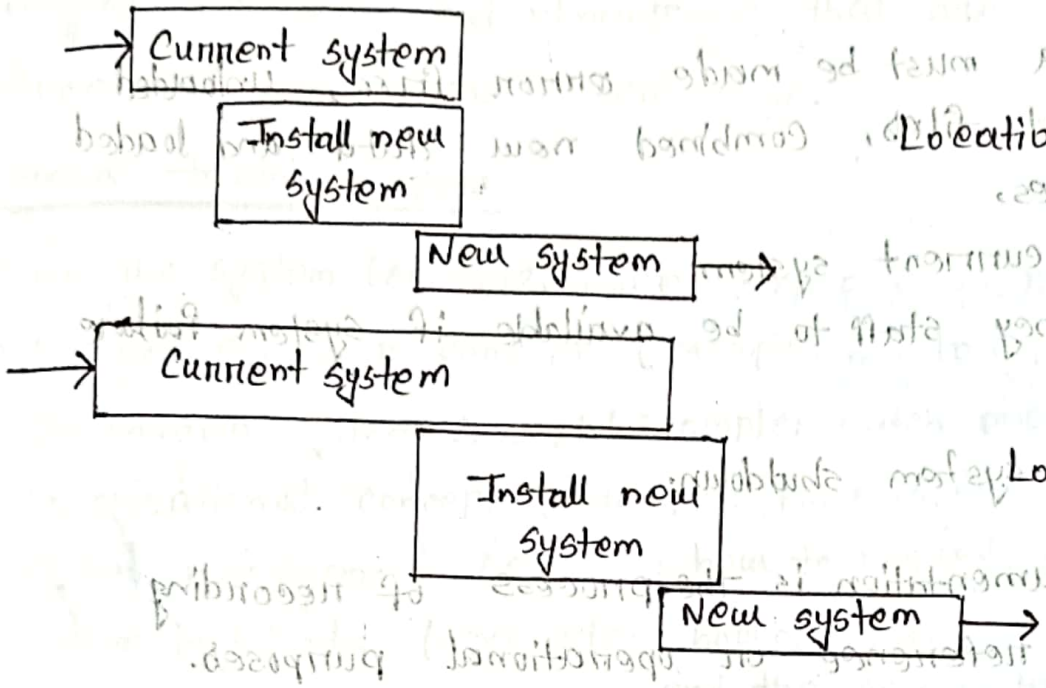


Parallel installation: Running the old information system and the new one at the same time until management decides the old system can be turned off.



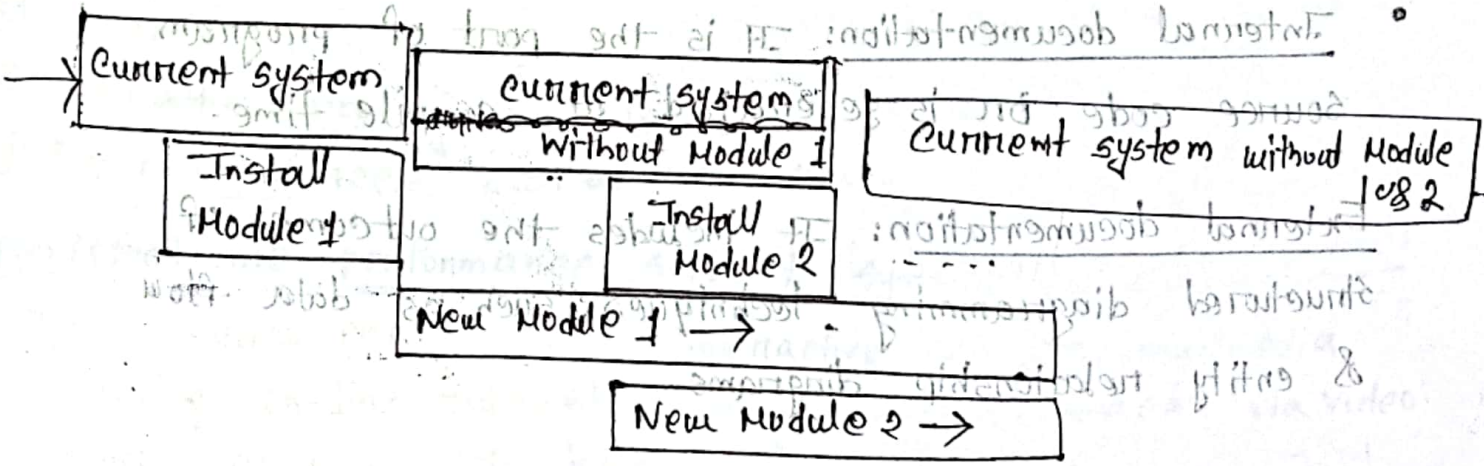
Single location installation: Trying out an information system at one site and using the experience to decide if and how the new system should be deployed throughout the organization.

Example: Install the system in the branch/smaller unit first



Phased installation: Changing the old information system to the new one incrementally, starting with one or a few functional components and then gradually extending the installation to cover the whole new system.

Example: Install the purchasing system first, then followed by inventory system and etc.



Planning installation:

⇒ Data conversion from existing system to new system.
⇒ Existing system contain data required by the new system.
⇒ Current data must be made error free, unloaded from current files, combined new data and loaded into new files,
⇒ Loading from current system.
⇒ Extra emergency staff to be available if system failure occurs
⇒ Planned old system shutdown.

Documentation: Documentation is the process of recording information for reference or operational purposed.

Types of documentation: ① System documentation
② User documentation

① **System documentation:** Detailed information about a system's design specifications, its internal workings and its functionality. It can be divided into two parts –

- **Internal documentation:** It is the part of program source code or is generated at compile time.
- **External documentation:** It includes the outcome of structured diagramming techniques such as data flow & entity relationship diagrams.

② User documentation: Written or other visual information about an application system how it works, and how to use it.

Training: Training in system analysis is the process of identifying the skills and knowledge that are needed to perform information systems tasks effectively.

Potential training topics:

① Use the system (example: how to enter a class registration request)
② General computer concept (example: how to copy the computer files)
③ Information system concept (example: Batch processing)
④ Organizational concept (example: First in first out – FIFO)
⑤ System management (example: how to request changes to a system)
⑥ System installation (example: how to ado adjustment current and the new system)

7 Training methods:

① Resident expert
② Computer aided instruction
③ Formal courses
④ Software help components
⑤ Tutorials
⑥ Interactive training manuals
⑦ External sources such as vendors

EPSS (Electronic performance support system:

⇒ Training modes include videos, interactive television, multimedia training, on-line tutorials. They maybe delivered via video tapes, CD-ROMs, the internet.

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⇒ EPSS is a component of a software package or application which training and educational information is embedded.

⇒ Reduce travel expenses and not required to send user off site for weeks.

⇒ Guide the user step by step.

Support information system:

It refers to the type of connection or constraint that a structure or a component has with its environment or other parts of the system. support is extremely important to users.

⇒ Providing support it can be expensive and time consuming.

Types of support system:

One approach is through automation,

① Internet based online support forums
② On-demand fax
③ Voice response systems
④ knowledge based (enables users access vendors knowledge bases including electronic support systems)

Providing support via help desk:

① Technical skill - expensive knowledge about how to use the system.
② People skills - Good listening and communication.

Support issues for the analyst to consider:

① Recovery and backup
② Disaster recovery
③ PC maintenance
④ Writing newsletters
⑤ setting up user groups

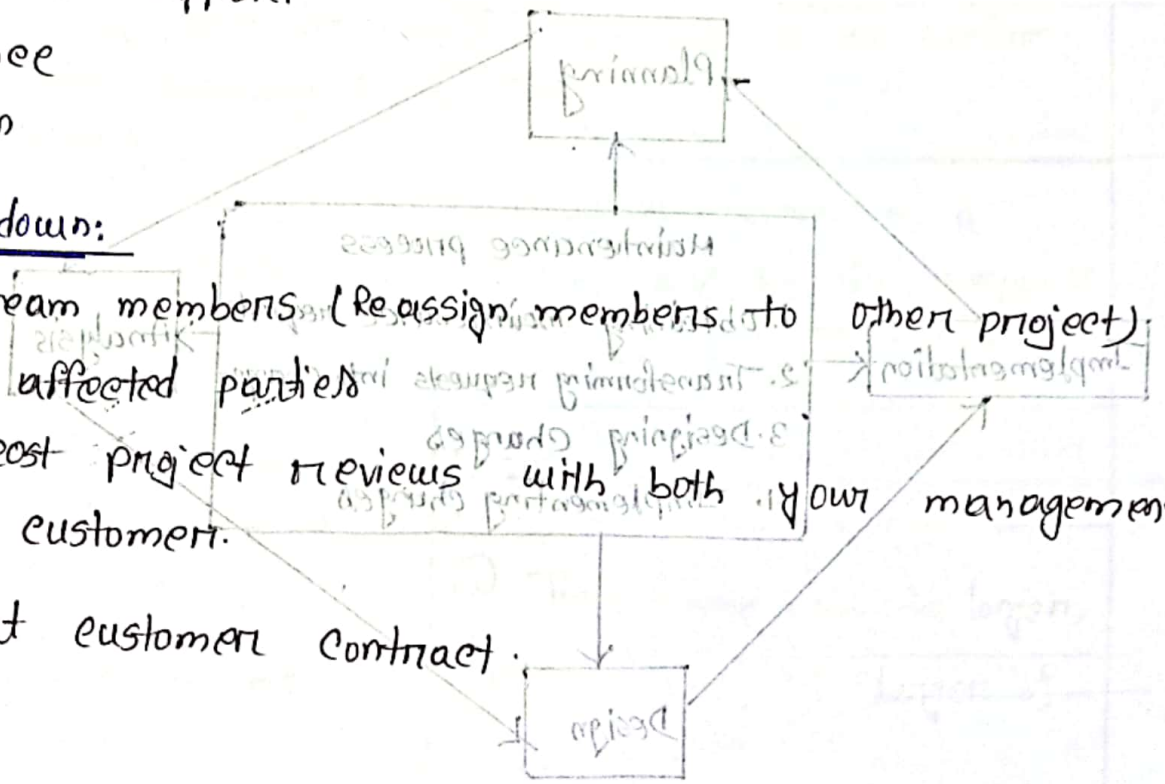
Implementation success factors:

- Major factors influencing the extent to which a system is used:

① personal stake of users (How relevant the system is to the work the user performs)
② System characteristics (ease of use, reliability, and relevance to the task)
③ user demographics (age, degree of computer experience)
④ Organizational support
⑤ performance
⑥ Satisfaction

Project close down:

⇒ Evaluate team members (Reassign members to other project)
⇒ Notify all affected parties
⇒ Conduct cost project reviews with both your management and your customer.
⇒ Close out customer contract.

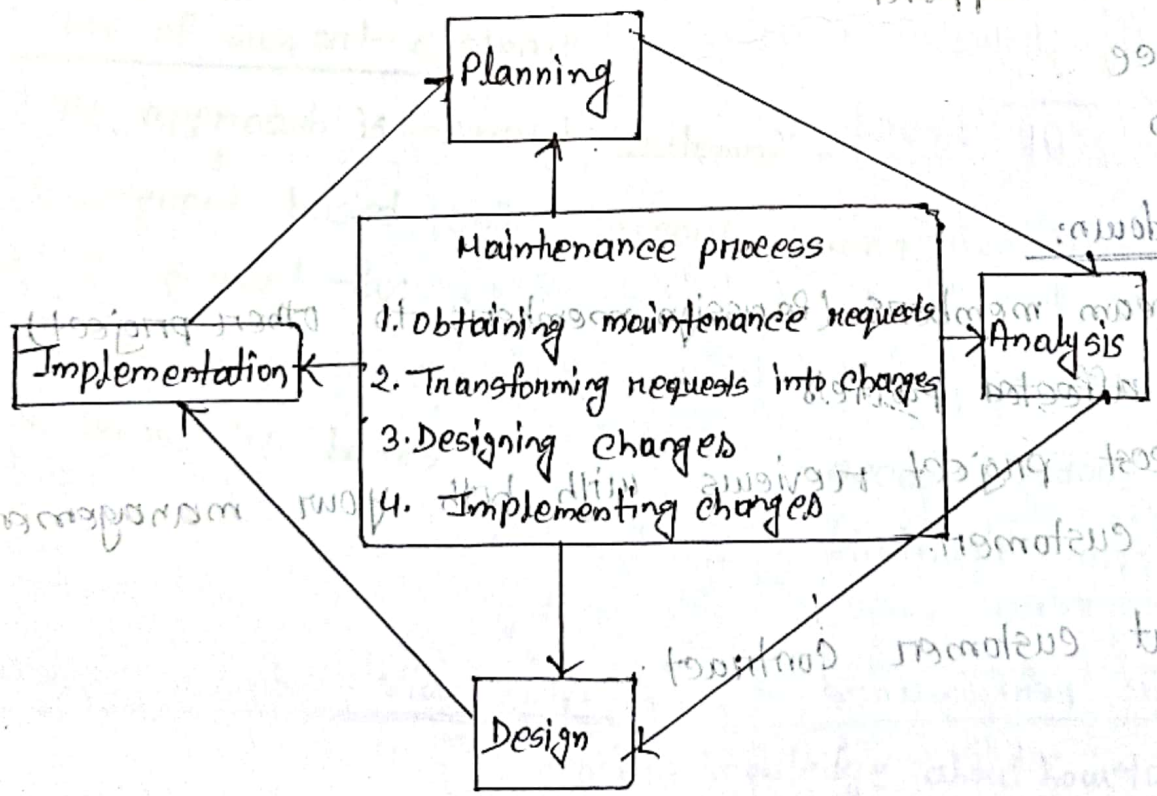


(Pre) **System maintenance:** Modification of a software product after delivery to correct faults, to improve performance or other attributes, or to adapt the product to a modified environment.

⇒ It is an ongoing activity which covers a wide variety of activities, including removing program and design errors updating documentation and test data and updating user support.

System maintenance activities: There are 4 activities:

① Obtaining maintenance requests
② Transforming requests into changes
③ Designing changes
④ Implementing changing



Types of system Maintenance:

① Adaptive: Modifying the system to cope with changes in the software ~~deve~~ environment (DBMS, OS)

② Perfective: Implementing new or changed user requirements which concern functional enhancements to the software.

③ Corrective: Diagnosing and fixing errors, possibly ones found by users.

④ Preventive: Increasing software maintainability or reliability to prevent problems in the future.

Corrective maintenance Vs adaptive maintenance;

Corrective maintenance	Adaptive maintenance
① Corrective maintenance is accomplished by removing bugs from software, hardware & network.	① Adaptive describe the modification of existing code to new platform.
② It is generally categorized into emergency situations such as a network outage or virus attack.	② Adaptive is used as a proactive and reactive response to changing business trends.
③ It's priority is high.	③ It's priority is low
④ Timeframe is typically short	④ Timeframe can be longer.
⑤ It's scope is small and simple	⑤ It's scope is larger of compl on.

Cost of maintenance: software maintenance cost is derived from the changes made to software after it has been delivered to the end user.

① Corrective – Generally 20% of software maintenance costs
② Adaptive – 25% of software maintenance costs.
③ Preventive – 5% of software maintenance costs
④ Perfective – 50% or more of software maintenance costs

Factors that influence system maintainability:

Software maintainability is defined as the degree to which an application is understood, repaired or enhanced.

① **Latent defects:** The number of unknown errors existing in the system after installation.

② **Number of customers for a given system:** Higher cost for higher number of users.

③ **Quality of system documentation:** Quality of technical system documentation including test cases.

④ **Maintenance personnel:** Number of quality of personnel dedicated to the support and maintenance of the system.

⑤ **Tools:** Software development tools, debuggers, hardware. Tools such as those that can automatically produce system documentation.

⑥ Software structure: Well structured programs make it easier to understand and fix programs.

Role of case tool in Maintenance:

Traditional systems development:

⇒ Emphasis on coding and testing
⇒ Changes implemented through coding and testing
⇒ Documentation follows maintenance
⇒ keeping documentation current is ignored for its time-consuming nature.

Development with case:

⇒ Emphasis is on design documents
⇒ Changes are implemented in design documents
⇒ Code is regenerated using code generators.
⇒ Documentation is updated during maintenance.

Website maintenance (special consideration):

⇒ 24 X 7 X 365 (?)
- The nature of continuous availability makes maintenance challenging.
- Pages under maintenance can be locked
- Date and time stamps may be included to indicate the most recent changes.

⇒ Check for broken links
⇒ HTML Validation
- New pages should be processed by a Code Validation routine before publication.

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⇒ Re-registration

- when content significantly changes, site may need to be re-registered with search engines.

⇒ Future editions

- Consistency is important to users.
- Post indications of future changes to the site.
- Batch changes to reduce the frequency of site changes.

Conducting system maintenance: three possible organizational structures:

① separate: Maintenence group consists of diffn personnel than development group.

② Combined: Developers also maintains systems.

③ Functional: Maintenance personnel work within the functional business unit.

Advantages: Separate - Improved system & documentation quality

Combined - Maintenance group knows all about system.

Functional - personnel have vested interested.

Disadvantages: Separate - Ignorance of critical undocumented information

Combined - Less emphasis on good documentation.

Functional - Limited job mobility & human on technical resources.

Conducting system maintenance (measures of effectiveness):

Important factor to consider

① Number of failures
② Time between each failure
③ Type of failure

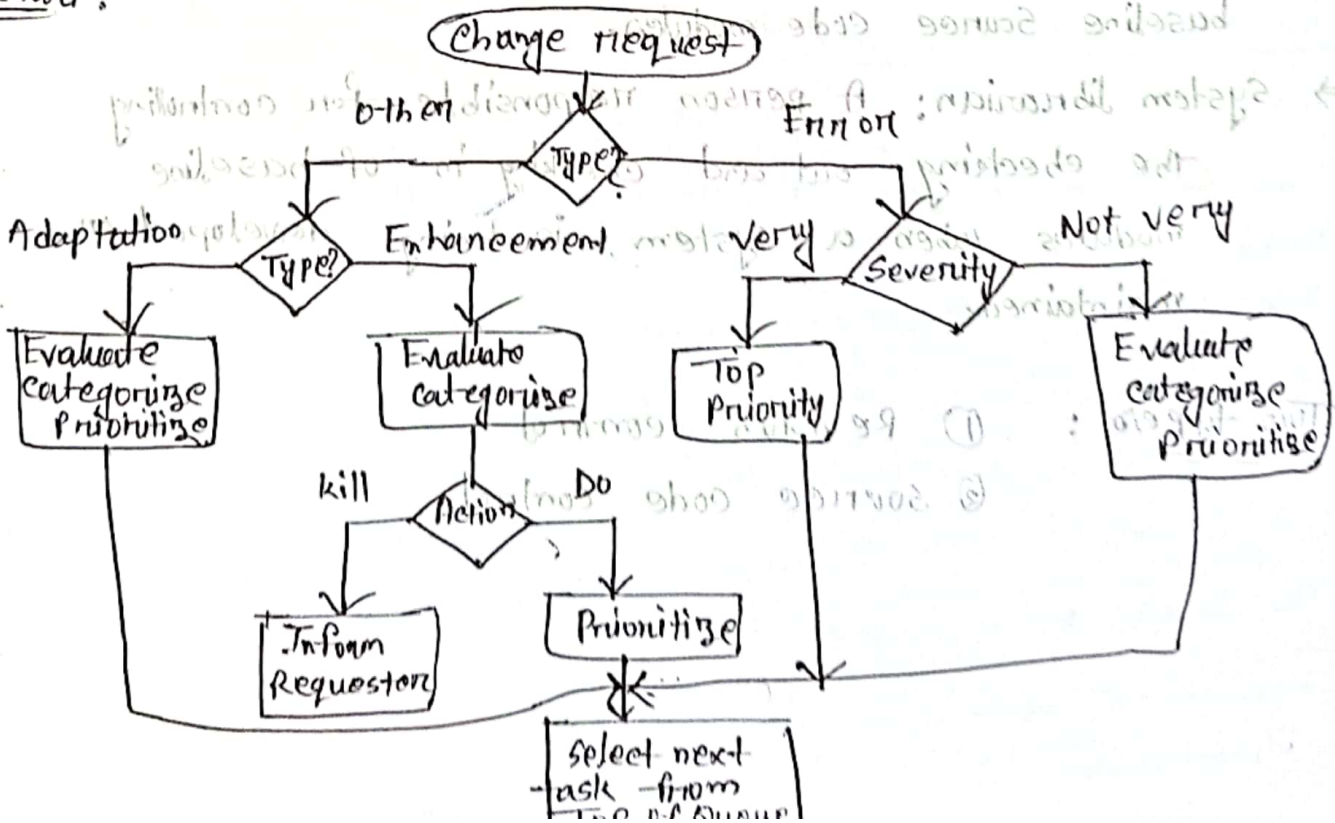
Mean time between failures (MTBF):

① It is the predicted elapsed time between inherent failures.
② It can be calculated as a arithmetic mean time between failures of a system.

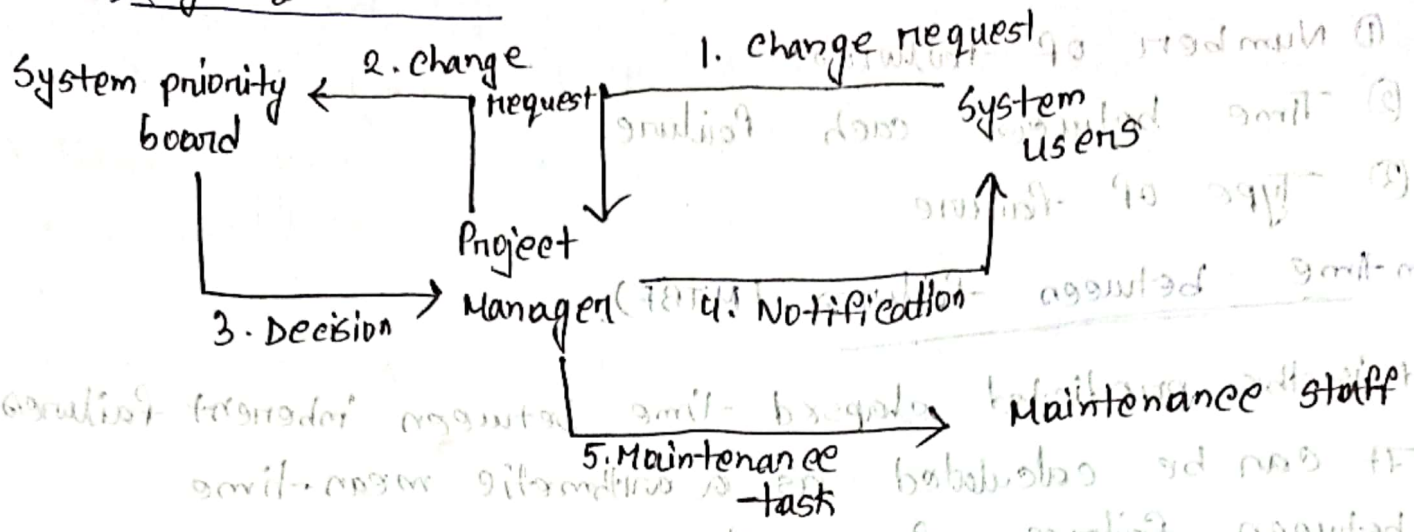
Controlling maintenance Request:

Possible types of request : ① Error - (corrective maintenance request)
② Adaptation
③ Enhancement

Flowchart:



④ How a maintenance request moves through an organization



Configuration management: It is the task of tracking & controlling changes in the software.

⇒ SCM can determine how to replicate it across many hosts.

⇒ Baseline modules: A system librarian controls the baseline source code modules.

⇒ System librarian: A person responsible for controlling the checking out and checking in of baseline modules when a system is being developed or maintained.

Two types: ① Revision control
② Source code control